



**FEDERAL UNIVERSITY OF BAHIA
SCHOOL OF ENGINEERING
DEPARTMENT OF STRUCTURAL AND
CONSTRUCTION ENGINEERING**



**LETTER OF SUPPORT
CEBE PhD Project**

September 23, 2026

To the CEBE Selection Committee

Dear Members of the Selection Committee,

I am pleased to express my support for the PhD project "Human-AI Collaborative Workflows for Immersive Digital Twins in Bridge Monitoring and Inspection," led by Aarhus University in collaboration with international research partners.

The project addresses an important challenge in the management of critical infrastructure by integrating Structural Health Monitoring, Digital Twins, semantic knowledge representation, multimodal artificial intelligence, and human-AI collaboration to support more informed and traceable infrastructure inspection and maintenance decisions.

I am pleased to support this research through collaboration with the project team and the PhD candidate. In particular, I intend to contribute expertise in structural engineering, infrastructure monitoring, and construction digitalization, and support the development of international case studies to evaluate the proposed methodology under real-world infrastructure conditions.

Subject to the necessary institutional approvals, permissions, safety requirements, and data-access agreements, I will support the identification of and facilitate access to suitable bridge and/or critical-infrastructure case studies in Brazil. These case studies can provide relevant inspection information, structural and environmental data, imagery, historical documentation, and other available evidence required for the development and validation of the proposed Digital Twin and AI-supported decision framework.

I am also willing to host the PhD candidate for a research stay in Brazil, providing an academic environment for collaboration with researchers and relevant infrastructure stakeholders. During this period, I will support the candidate in case-study definition, field activities, interpretation of local infrastructure conditions, and evaluation of the transferability of the developed methodology.

The Brazilian case studies are particularly relevant because they will enable the project to investigate the framework's applicability under infrastructure-management conditions that may differ substantially from those encountered in Denmark and Switzerland, including differences in climate, monitoring availability, historical information, and inspection and maintenance resources.

I look forward to collaborating with Aarhus University and the other project partners and fully support this PhD application.

Sincerely,



Professor Dayana Bastos Costa
Federal University of Bahia (UFBA)
Salvador, Bahia, Brazil